

SECTION 1 – INTRODUCTION

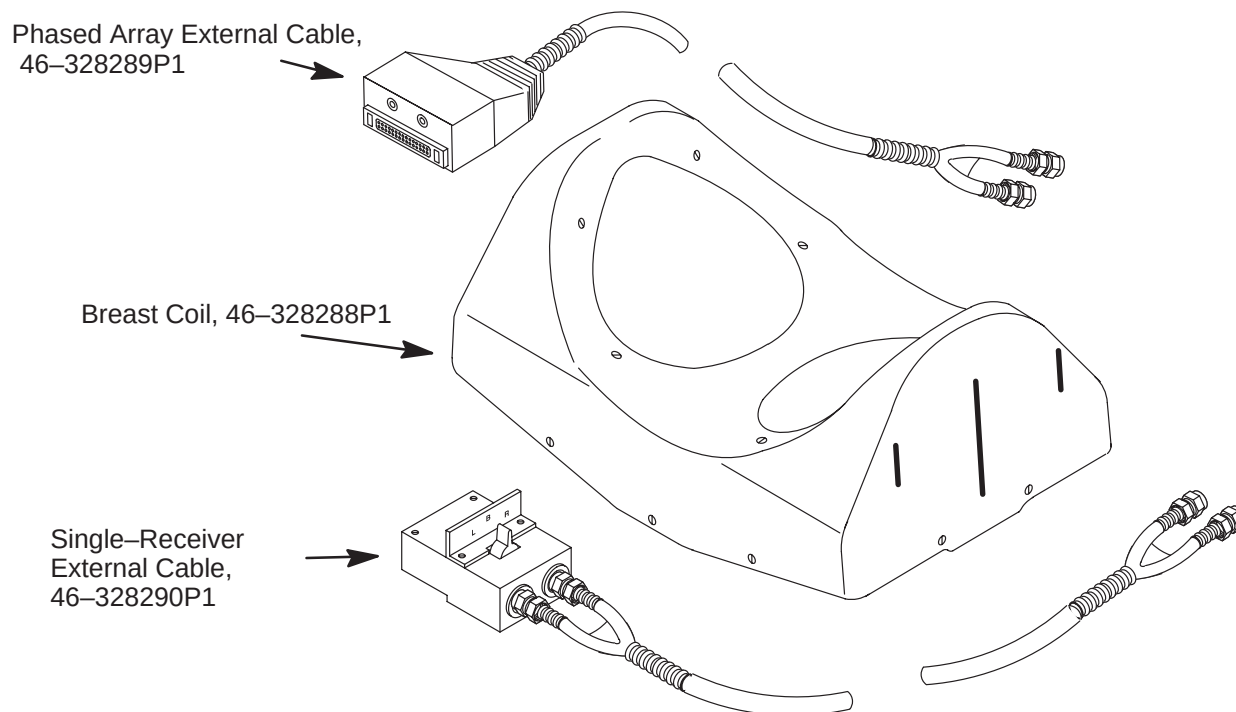
1-1 HOW THE BREAST COIL OPERATES

The Breast coil is a two element receive only antenna array which receives the RF signal from the body coil and sends it to the preamplifier. The coil is used to image the female breast. The coil has two main applications. First, the detection of breast pathology, in particular, breast cancer. Second, the examination of silicon breast implants.

The Breast coil can be used in two configurations (single-receiver or phased array) determined solely by which external cable is used. The coil is shipped with the single-receiver cable installed. A switch on the single-receiver cable allows the Breast coil to be used in three modes: unilateral left, bilateral, or unilateral right. A phased array cable is also shipped with each coil. If the coil is intended to be used as a phased array coil, or if the site purchases the phased array option in the future, the single-receiver coil cable can be removed and the phased array cable installed. Do not throw the extra cable away!

The small individual antenna loop elements (2 in total) provide excellent SNR across a 32cm FOV. The Breast coil is tuned to the 0.5T MR frequency, 21.29 MHz.

The Coil consists of three major assemblies: (See Illustration 1-1)



BREAST COIL ASSEMBLIES

ILLUSTRATION 1-1

1–1 HOW THE BREAST COIL OPERATES (continued)

The two coil element assemblies each contain a printed Kapton conductor mated with a circuit board containing the matching and decoupling components. A tuning capacitor is included at the coil interface printed circuit board to enable the coil to be adjusted to exact resonance at 21.29 MHz. The combined series value of the tuning, decoupling, and drive capacitors on the interface printed circuit board was selected to resonate the coil at 21.29 MHz. The ratio of the decoupling and tuning capacitor was selected to provide a matched output impedance of 50 ohms under nominal loading conditions. The drive capacitor is actually composed of two capacitors in a parallel configuration to provide increased RF voltage rating for this component without resorting to a custom high voltage RF capacitor. See Illustration 1–2.

An inductor in series with the coil output allows preamplifier activated multicoil decoupling to be accomplished by the low shunt impedance of the preamplifier input. A PIN diode is installed at this location to aid in transmit decoupling, provide a proper load for the T/R bias driver, and protect the preamplifier. The inductor is adjusted to resonate with the drive capacitor at 21.29 MHz, thus inserting a high impedance in series with the coil loop at the transmit RF field frequency whenever the PIN diode is biased into the ON state. In a similar manner, a low impedance at the coil output (from the input of the multicoil array (MCA) preamplifiers) will cause one element of the coil system to decouple from the other to provide coil element-to-coil element isolation in MCA mode operation. Because of the coil element positions dictated by the anatomy, no significant isolation from overlap of the two coil elements was possible. The output of the coil is terminated in an SMA connector to enable the output cable to be readily changed.

A passive decoupling circuit consisting of a group of four crossed diodes and a series inductor are shunted across the decoupling capacitor. The inductor is tuned to resonate with the decoupling capacitor at 21.29 MHz when the crossed switching diodes are forward biased by induced RF power from the transmit excitation field. The added passive decoupling circuit is used to ensure a significant margin of safety in achieving sufficient transmit decoupling. Because of the very large value of capacitor required at the drive point, the decoupling impedance of the network at this location is not sufficient to provide the desired safety margin.

In order to allow the preamplifier input impedance to effectively isolate one coil from the other, the electrical length from the shunt PIN diode on the output of the coil electronics to the preamplifier input must be either 0, or an exact multiple of one-half wavelength. At 0.5T, the cable would need to be nearly 180 inches long. In order to maintain the cable length used at 1.5T, a Pi section network with a characteristic impedance of 50 ohms and an electrical length of 60° is used at the output of each coil, located on the interface printed circuit board.

The cable set for the MCA mode consists of a dual cable having two SMA terminations at the coil end, and a 30 pin connector block designed to mate with the MCA coil port on the host system. The output of the left coil element is connected to MCA coil port 2, corresponding to system receiver 1. The output of the right coil element is directed to MCA coil port 5, corresponding to system receiver 0. The physical length of the cable has been carefully cancelled to 0° by the Pi section network on the coil element to the input of the MCA preamplifier. This ensures that the low input impedance of the MCA preamplifier provides some isolation between the two coil elements via a decoupling action.

The cable set for the non-MCA mode consists of dual cable having two SMA terminations at the coil end, a switch to deselect one coil element when a unilateral scan having optimum signal to noise ratio is desired, a combiner to allow bilateral signal acquisitions, and a 20 pin connector to mate with the host system head/surface coil port.

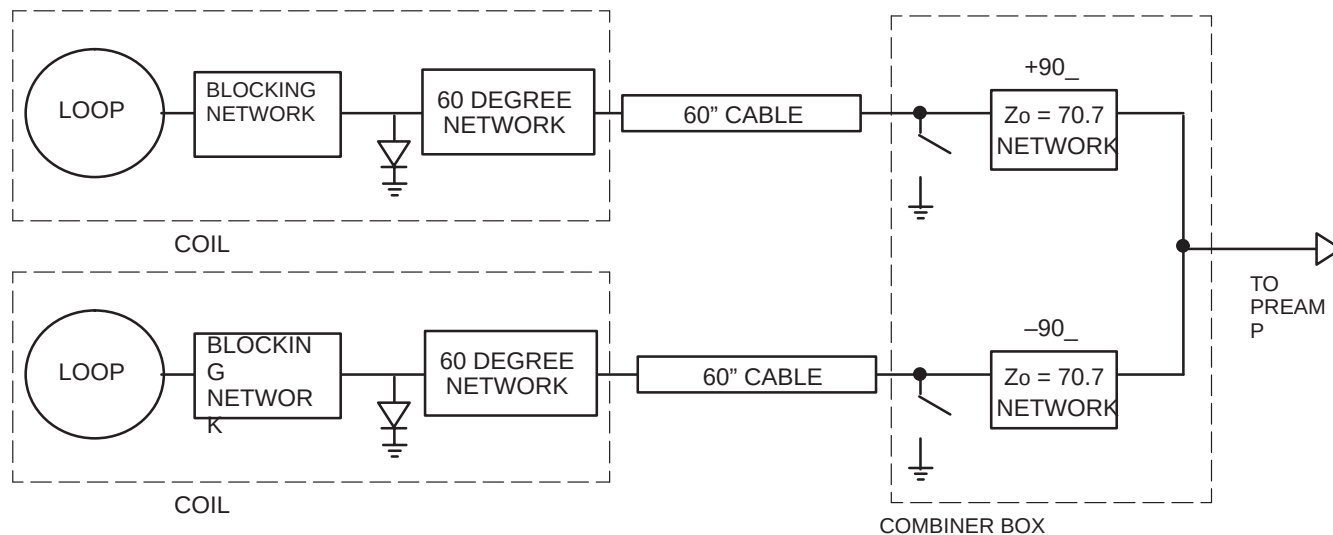
1–1 HOW THE BREAST COIL OPERATES (continued)

The combiner consists of two 90 degree 70.7 ohm Pi sections, one having a positive phase shift, and the other having a negative phase shift. As the output components from the two desired networks are of equal reactance but of opposite sign, they are eliminated from the final design. When combined, the signals from the two coil elements are added with a net phase shift of 180 degrees. This is necessary because the two coil elements are identical in design and configuration, but assembled with one coil element rotated 180 degrees relative to the other. As the 50 ohm impedance of each coil element is seen by the system preamplifier through a 70.7 ohm, 90 degree network, the preamplifier sees two 100 ohm sources in parallel, resulting in the desired 50 ohm source impedance.

When a unilateral acquisition is desired, the selector switch is used to short out the output from the undesired coil. This has two effects. First, as the cable from the coil element to the switch, in combination with the Pi network on the coil interface card is exactly $0\lambda/4$ long electrically, the short created by the switch has an effect similar to biasing on the PIN diode of the unused coil element, thus decoupling it from the desired coil during data acquisition. As the short is an RF short only, the DC bias for the PIN diode is still supplied as under normal imaging. Thus the transmit mode decoupling is unaffected. Second, the switch shorts the unused input to the combiner, causing the Pi section to invert the short into an open at the preamplifier port end of the network. As a result, the signal from the desired coil passes through a single Pi section, resulting in a minor mismatch (50 ohm impedance of the coil will be transformed to 100 ohms by the 70.7 ohm Pi network), but in no other loss of signal.

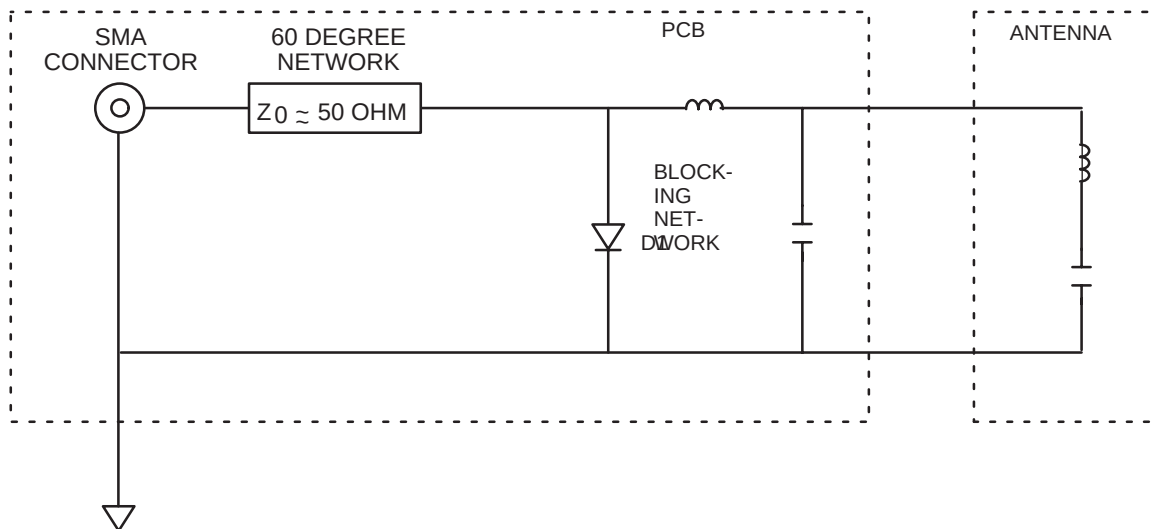
The PIN diode bias for the left side coil element is obtained from the system receive port bias driver in the conventional manner, via the series inductor in the combiner 90 degree network. To ensure the optimum level of transmit field decoupling, the bias for the right side coil element is obtained via an RF isolation network from the transmit port bias driver of the system head/surface coil port. The path is through a 38 μ H RF choke, and then the shunt inductor of the combiner 90 degree network.

1-1 HOW THE BREAST COIL OPERATES (continued)



BLOCK DIAGRAM FOR SINGLE-RECEIVER CONFIGURATION
ILLUSTRATION 1-2

There are two circuit boards and two antennas in each Breast coil. See Illustration 1-3 for a simple block diagram.



BLOCK DIAGRAM FOR SINGLE COIL
ILLUSTRATION 1-3

1-2 COMPATIBILITY

The Breast Array Coil is compatible with the following hardware configurations:

- Signa Advantage™ 0.5T (0.5T only)

1-3 RELATED DOCUMENTS

- *Direction 15400, Signa Advantage™ 1.5T, 1.0T, & 0.5T System*

1-4 ORGANIZATION OF THIS DOCUMENT

This document is divided into the following sections:

Section 1, INTRODUCTION, describes how the Breast Coil operates and when and where the Breast Coil can be used.

Section 2, SET UP AND CALIBRATION, describes installation procedures.

Section 3, FUNCTIONAL CHECKS, describes the normal power-up sequence.

Section 4, REPLACEMENT/MAINTENANCE, describes field maintenance procedures.

Section 5, RENEWAL PARTS, lists field replaceable parts.

Appendix A, SCHEMATICS, assembly drawings and schematics.

1-5 ENVIRONMENTAL REQUIREMENTS

Operate and store the Breast Coil in the Scanner Room.

SECTION 2 – SET UP AND CALIBRATION

2-1 CHECKING THE SHIPPING LIST

Table 2-1 lists the M1685BR Signa 0.5T Breast Coil parts. Check that all parts have been shipped.

TABLE 2-1
BREAST COIL SHIPPING LIST (46–328198G1)

QTY.	ITEM	PART NO.
1	Bilateral Breast Coil	46–328288P1
1	Single–Receiver Cable	46–328290P1
1	Phased Array Cable Repair Kit	46–328289P1
2	Arm Pad	N/A
1	Chest Pad	N/A
2	A–sized (3") Filler Pad	N/A
2	B–sized (2") Filler Pad	N/A
2	C–sized (1") Filler Pad	N/A
1	Abdomen Pad	N/A
1	Foot Pad	N/A
1	Face Pad	N/A
2	Cradle Attachment Strap	N/A
5	Disposable Liner, Chest	N/A
5	Disposable Liner, Face	N/A
1	Operator Manual	N/A
1	Service Manual	46–305161

2-2 INSTALLING THE BREAST COIL

At the Operator Console, install new soft key to be used by the operator to select Breast imaging. Name the key "BREAST" for Breast single–receiver coil or "BREASTPA" for Phased Array Breast Coil. Some proprietary software tools key off of the coil name, use the correct coil name to ensure these tools function.

Refer to the *Direction 15400, Signa Advantage 1.5T, 1.0T, & 0.5T System*, Set–up & Calibration tab, Section 6, CONFIGURATION FILE for information on installing soft keys and configuring the Breast Coil.

The Breast Coil is delivered with the single–receiver cable installed. Refer to Section 4–2, REPLACING THE EXTERNAL CABLE, for installing the Phased Array cable (46–328289P1).

2-3 FUNCTIONAL CHECKS

1. Perform a Body coil scan SNR verification. Refer to Section 3–1, BODY COIL SNR VERIFICATION.
2. Perform a Breast coil SNR Verification. Refer to Section 3–2, BREAST COIL SNR VERIFICATION.
3. Perform Spike Noise Analysis for the Breast Coil. Refer to *Direction 15400, Signa Advantage™ 1.5T, 1.0T, & 0.5T System*, Performance Checks tab, Section 3, SPIKE NOISE.

2-4 PERIODIC QUALITY ASSURANCE CHECK

On a periodic basis, such as during planned maintenance, perform the quality assurance check outlined below to ensure that the Breast coil is operating properly with no appreciable degradation of image quality.

1. Perform a Breast coil SNR Verification. Refer to Section 3–2, BREAST COIL SNR VERIFICATION.
2. Record the date and value calculated in Section 3–3, SNR IMAGE ANALYSIS under “SNR Data QA Check” of the Data Table as is instructed.
3. As is instructed in the Data Table, divide the SNR value obtained in the periodic QA check by the original SNR value.
4. If this ratio is not greater than 85%, then there may be a problem with the Breast coil of the MR system. Contact your local GE Service Representative.

SECTION 3 – FUNCTIONAL CHECKS

3-1 BODY COIL SNR VERIFICATION

Phantom Required

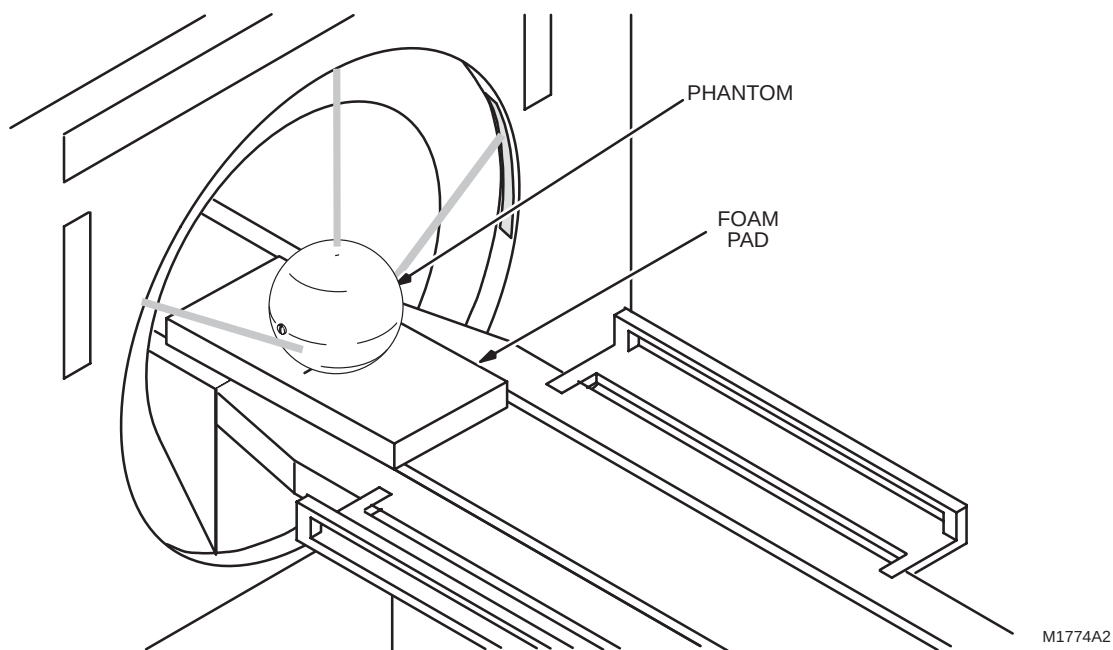
- 100 mm TLT Phantom, 46–317586G1

Setup Procedure



The Quad Head Coil must be completely removed from the cradle before performing any body scans. Failure to do this may result in damage to the Head Coil T/R Network.

1. Remove Quad Head Coil (if present) from cradle.
2. Select **[New Exam]** (5.x) to allow a new Landmark to be set.
3. Position the 100 mm TLT phantom in the center of the cradle. Use foam pads to raise the phantom to the same height as in the Breast Coil. Landmark the center of the phantom and advance the phantom to isocenter using the **[ADVANCE TO SCAN]** button. See Illustration 3–1.



100 MM TLT PHANTOM LANDMARK SETUP FOR BODY COIL SNR
ILLUSTRATION 3–1

3–1 BODY COIL SNR VERIFICATION (continued)

4. Setup Scan Prescription as shown in Table 3–1 (5.x).

TABLE 3–1
BODY COIL SNR – SCAN PROTOCOL (5.x)

SCAN PRESCRIPTION (5.x)	
<u>MAIN MENU</u>	[New Exam]
<u>PATIENT/EXAM INFORMATION</u>	
ID:	geservice
Name:	body snr
Weight:	300
	[Patient Position]
<u>PATIENT POSITION</u>	
Patient Entry	[Head First]
Patient Position	[Supine]
Axial/Sag. Landmark	[Sternal Notch]
Coil Type	[Body Coil]
Scan Plane	[Sagittal]
	[Imaging Params]
<u>IMAGING PARAMETERS</u>	
Image Mode	[2D]
(K SAR must be "On" K)	[Monitor SAR] (Release 5.2)
Pulse Sequence	[Spin Echo]
Imaging Option	[None]
Enter PSD Filename:	
Monitor SAR?	[Yes] (Release 5.1)
	[Next Screen] [Scan Timing]
<u>SCAN TIMING</u>	
Number of Echoes	[1]
Echo Time (TE)	[Other] 40 msec
Rep Time (TR)	[Other] 400 msec
	[Scanning Range]
<u>SCANNING RANGE</u>	
Field of View	[Other] 32 cm
Scan Thickness	[3 mm]
Interscan Spacing	[Other] 0
Start Loc: (I/S)	0
End Loc: (I/S)	0
No. of Scan locations	1
FOV Center: (L/R)	0 (P/A) 0
	[↩] [Acq Time]
<u>ACQUISITION TIME</u>	
Acq. Matrix (freq.)	[256] (Release 5.2)
Acq. Matrix (phase)	[256] (Release 5.2)
Acquisition Matrix	[256 x 256] (Release 5.1)
Frequency Direction	[A/P]
Imaging Time	[1 NEX 1:49]
Contrast	[NO]
Table Delta:	0 mm
	[Scan Set-up]
<u>SCAN SET-UP</u>	
Auto CF	[Peak]
Primary Arch. Node	Default
	[Scan Ops]
<u>SCAN OPERATIONS</u>	

5. Select **[Auto Prescan]** to properly calibrate the RF power level for the 90 degree and 180 degree pulses.
6. Select **[Scan]**. Observe the resulting image. Ensure that there are no artifacts of any sort in the resulting image. Record the Exam number and Series number for SNR Calculations.
7. Select **[Scan]** again. This second image will be used for determination of Body Coil mode SNR.
8. After completion of scan, select **[Cancel]**. Refer to Section 3–3 for SNR image analysis.

3-2 BREAST COIL SNR VERIFICATION

Note

An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to "TLT PROCEDURE" calibration procedure located in Direction 15491, *Signa Advantage 1.5T, 1.0T, & 0.5T System Troubleshooting Guide*.

Phantom Required

- 100 mm TLT Phantoms (2), 46–317586G1

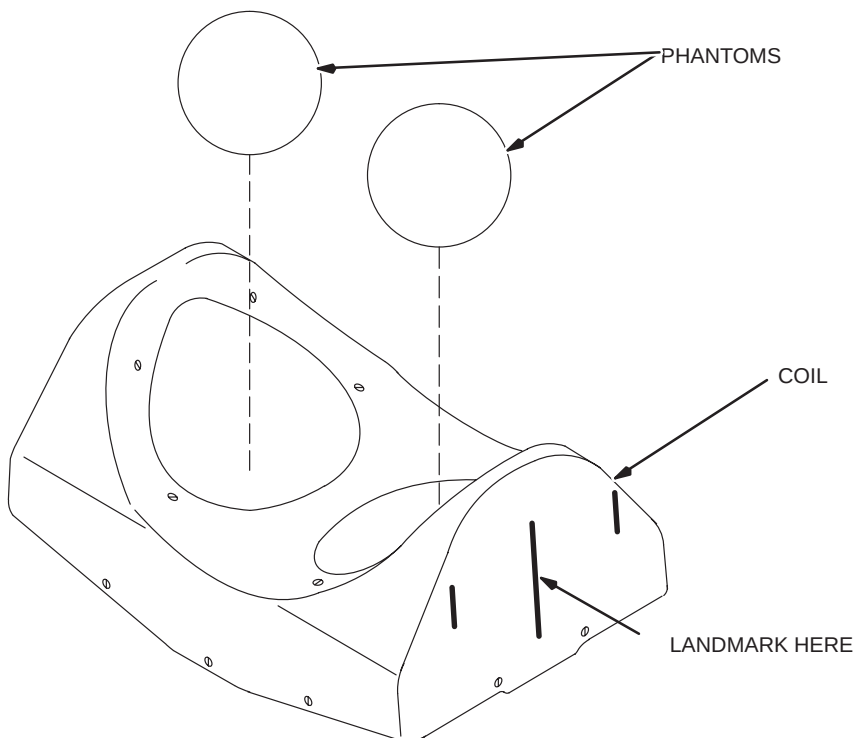
Setup Procedure

1. Select **[New Exam]** (5.x) to allow a new Landmark to be set.



The Quad Head Coil must be completely removed from the cradle before performing any body scans. Failure to do this may result in damage to the Head Coil T/R Network.

2. Remove Quad Head Coil (if present) from cradle.
3. Place Breast Coil to be tested onto table.
4. Connect Breast Coil connector to its mating connector in the Carriage Assembly.
5. Place the one phantom in each Breast Coil well. See Illustration 3–2.
6. Landmark using vertical line on side of Breast Coil and advance the coil to isocenter.



BREAST COIL PHANTOM SETUP
ILLUSTRATION 3–2

3–2 BREAST COIL SNR VERIFICATION (continued)

7. Setup Scan Prescription as shown in Table 3–2 (5.x).

TABLE 3–2
BREAST COIL SNR – SCAN PROTOCOL (5.x)

SCAN PRESCRIPTION (5.x)	
<u>MAIN MENU</u> [New Exam] <u>PATIENT/EXAM INFORMATION</u> ID: geservice Name: body snr Weight: 300 [Patient Position] <u>PATIENT POSITION</u> Patient Entry [Head First] Patient Position [Supine] Axial/Sag. Landmark [Sternal Notch] Coil Type [Other Coils] [BREAST] Note 1 Scan Plane [Axial] [Imaging Params] <u>IMAGING PARAMETERS</u> Image Mode [2D] (K SAR must be "On" K) [Monitor SAR] (Release 5.2) Pulse Sequence [Spin Echo] Imaging Option [None] Enter PSD Filename: Monitor SAR? [Yes] (Release 5.1) [Next Screen] [Scan Timing] <u>SCAN TIMING</u> Number of Echoes [1] Echo Time (TE) [Other] 40 msec Rep Time (TR) [Other] 400 msec [Scanning Range]	<u>SCANNING RANGE</u> Field of View [Other] 32 cm Scan Thickness [3 mm] Interscan Spacing [Other] 0 Start Loc: (I/S) 0 End Loc: (I/S) 0 No. of Scan locations 1 FOV Center: (L/R) 0 (P/A) 0 [↩] [Acq Time] <u>ACQUISITION TIME</u> Acq. Matrix (freq.) [256] (Release 5.2) Acq. Matrix (phase) [256] (Release 5.2) Acquisition Matrix [256 x 256] (Release 5.1) Frequency Direction [R/L] Imaging Time [1 NEX 1:49] Contrast [NO] Table Delta: 0 mm [Scan Set-up] <u>SCAN SET-UP</u> Auto CF [Peak] Primary Arch. Node Default [Scan Ops] <u>SCAN OPERATIONS</u>

Note 1: Select **[BREASTPA]** for phased array configuration of Breast Coil.

8. Select **[Auto Prescan]** to properly calibrate the RF power level for the 90 degree and 180 degree pulses.
9. Select **[Scan]**. Observe the resulting image of the spheres. See Illustration 3–3. Ensure that there are no artifacts of any sort in the sphere images. Record the Exam number and Series number for SNR Calculations.
10. Select **[Scan]** again. This second image of the spheres will be used for determination of BREAST mode SNR.
11. After scan is complete, select **[Cancel]**. Check image for indications of coupling, see Illustration 3–3. Refer to Section 3–3 for SNR image analysis.

3-2 BREAST COIL SNR VERIFICATION (continued)

PROPERLY DE-COUPLED IMAGE



IMPROPERLY COUPLED IMAGE

NOTE: IF COUPLING IS PRESENT, NOTIFY YOUR GEMS SERVICE REPRESENTATIVE.

BREAST COIL IMAGES

ILLUSTRATION 3-3

3-3 SNR IMAGE ANALYSIS (RELEASE 5.x)

Description

The SNR tool retrieves two operator selected images. Signal value is computed as the mean pixel value in a ROI covering 80% of the image. The image is analyzed to determine the center of the image for positioning the ROI. A difference image is created by subtracting the second image from the first and the same ROI is used to calculate noise from the subtracted image. The signal value, noise value, and signal to noise ratio are reported. There is an option to save the difference image with the results annotated.

Procedure

1. Touch **[UTILITIES]**, **[MR Tools]**, **[Image Quality]**, then **[SNR Test]**. The SNR Test screen is displayed; see Illustration 3-4.
2. Enter first image exam, series, and image numbers. If exam, series, or image numbers are not known, select **[List Exams]**, **[List Series]**, or **[List Images]** to display list to choose from.

Note

Image number selection must be back lit (highlighted) to be able to enter information. Use Switch key on keyboard to transfer control from left to right side of Touch Screen.

3. Enter second image series and image numbers. If series or image numbers are not known, select **[List Series]** or **[List Images]** to display list to choose from. The second image Exam Number is defaulted to the same as first image.

NEMA STANDARD SNR TEST

Filter
Off

Image
Off

First
Image

List/Sel
Exams
xxx

List/Sel
Series
x

List/Sel
Images
x

Second
Image

List/Sel
Series
x

List/Sel
Images
x

Body Axial SNR Data

Signal : xx.x

Noise : xx.x

SNR : xx.x

Cancel

Utility
Main

MR Tools
Main

Image
Quality

Accept



Note: Accept changes to continue only after an analysis has been performed.

SNR TEST SCREEN

ILLUSTRATION 3-4

3–3 SNR IMAGE ANALYSIS (RELEASE 5.x) (continued)

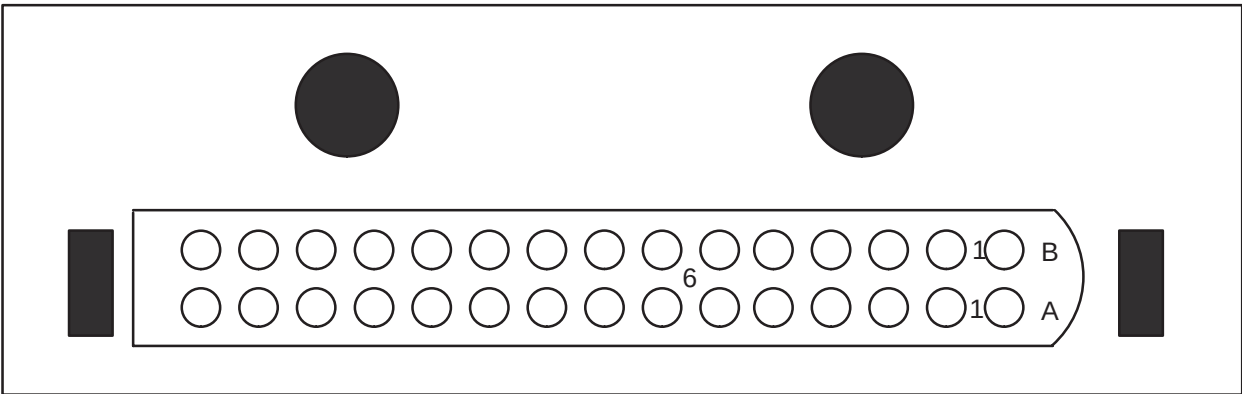
4. If high pass filtering is desired to be performed on data, touch **[Filter Off]** which will highlight and change to **[Filter On]**.
5. If the difference image annotated with data is to be created, touch **[Image Off]** which will highlight and change to **[Image On]**.
6. Touch **[Accept]** to begin analysis. The final values are displayed on the touch screen, see Illustration 3–4.
7. Touch **[Continue]** then select the next exam and repeat the above analysis for each image pair.
8. Record the date and value calculated in the appropriate column under “SNR Data QA Check” of the Data Table on page DT–1 (following Section 5, RENEWAL PARTS) as is instructed.

3-4 CHECKING THE PROTECTION PIN DIODES WITH DIGITAL MULTIMETER (DMM)

- 1. Select the DIODE TEST function on the Digital Multimeter (DMM).
- 2. Connect the NEGATIVE lead of the DMM to the external cable connector row A. Connect the POSITIVE lead to the external cable connector row B. Refer to Table 3–3 and Illustration 3–5 (Phased Array) or Table 3–4 and Illustration 3–6 (Single–Receiver).
- 3. A reading of 0.400 to 0.900 should be observed on the DMM.
- 4. If a reading below 0.400 is observed in either direction, either the output cable is shorted or bad PIN diode.
- 5. If a reading above 0.900 is observed in step 2, the PIN diode is defective.
- 6. Connect the NEGATIVE lead of the DMM to the external cable connector row B. Connect the POSITIVE lead to the external cable connector row A. Refer to Table 3–3 and Illustration 3–5 (Phased Array) or Table 3–4 and Illustration 3–6 (Single–Receiver).
- 7. A reading of INFINITY should be observed on the DMM.
- 8. If a reading of INFINITY is observed in both directions, either the output cable is open or the PIN diode is open.
- 9. If any of the above conditions fails, replace the PIN diode. Refer to Section 4–3, REPLACING THE PROTECTION/DE–COUPLING PIN DIODES.

TABLE 3–3
DIODE TEST CONNECTIONS (phased array COIL)

COIL NUMBER	POSITIVE LEAD CONNECTION		NEGATIVE LEAD CONNECTION	
	FOR STEP 2	FOR STEP 6	FOR STEP 2	FOR STEP 6
1	A3 or B3	A2 or B2	A2 or B2	A3 or B3
2	A9 or B9	A8 or B8	A8 or B8	A9 or B9

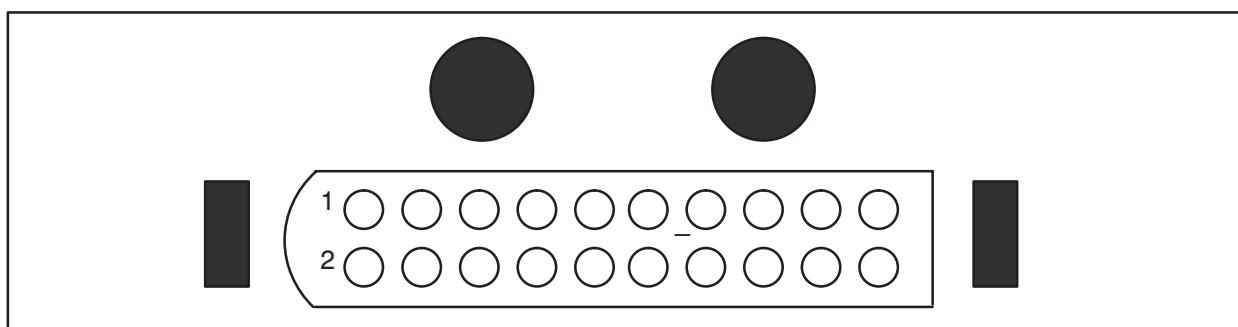


PIN LOCATIONS FOR PHASED ARRAY CONNECTOR
ILLUSTRATION 3–5

3-4 CHECKING THE PROTECTION PIN DIODES WITH DIGITAL MULTIMETER (DMM) (continued)

TABLE 3-4
DIODE TEST CONNECTIONS (single-receiver)

COIL NUMBER	POSITIVE LEAD CONNECTION		NEGATIVE LEAD CONNECTION	
	FOR STEP 2	FOR STEP 6	FOR STEP 2	FOR STEP 6
1	B2	A6	A6	B2
2	B9	A6	A6	B9



PIN LOCATIONS FOR SINGLE-RECEIVER CONNECTOR

ILLUSTRATION 3-6

3-5 CHECKING THE EXTERNAL CABLE

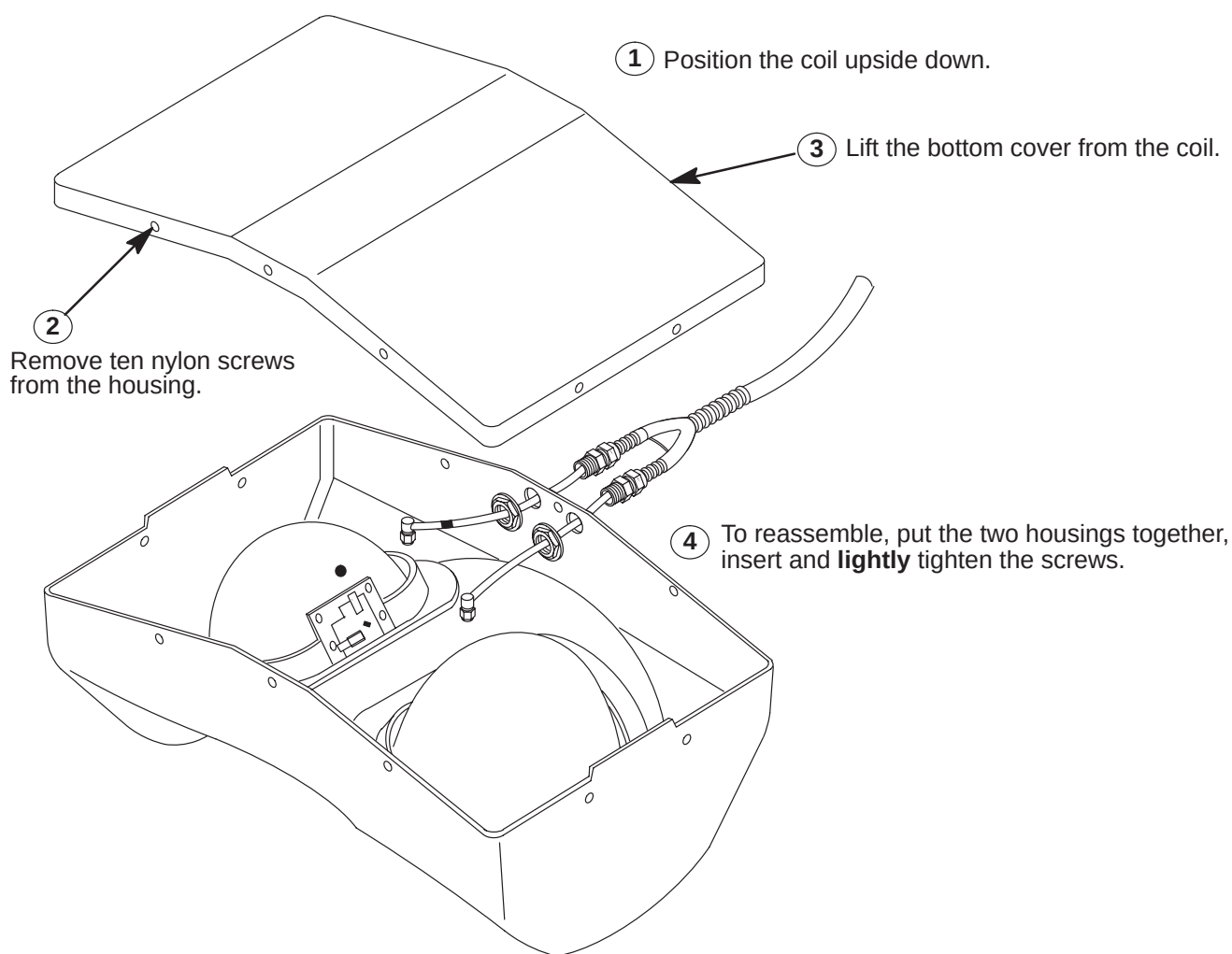
1. Select the DIODE TEST function on the Digital Multimeter (DMM).
2. Connect the NEGATIVE lead of the DMM to the external cable connector row A. Connect the POSITIVE lead to the external cable connector row B. Refer to Table 3-3 and Illustration 3-5 (Phased Array) or Table 3-4 and Illustration 3-6 (Single-Receiver).
3. Flex the external cable, especially near the connectors and the strain relief, and observe that a reading of 0.400 to 0.900 should remain on the DMM, with no instability or fluctuations.
4. Connect the NEGATIVE lead of the DMM to the external cable connector row B. Connect the POSITIVE lead to the external cable connector row A. Refer to Table 3-3 and Illustration 3-5 (Phased Array) or Table 3-4 and Illustration 3-6 (Single-Receiver).
5. Flex the external cable, especially near the connectors and the strain relief, and observe that a reading of INFINITY should remain on the DMM, with no instability or fluctuations.
6. If the cable fails any of the above tests, it should be replaced. Refer to Section 4-2, REPLACING THE EXTERNAL CABLE.

Blank

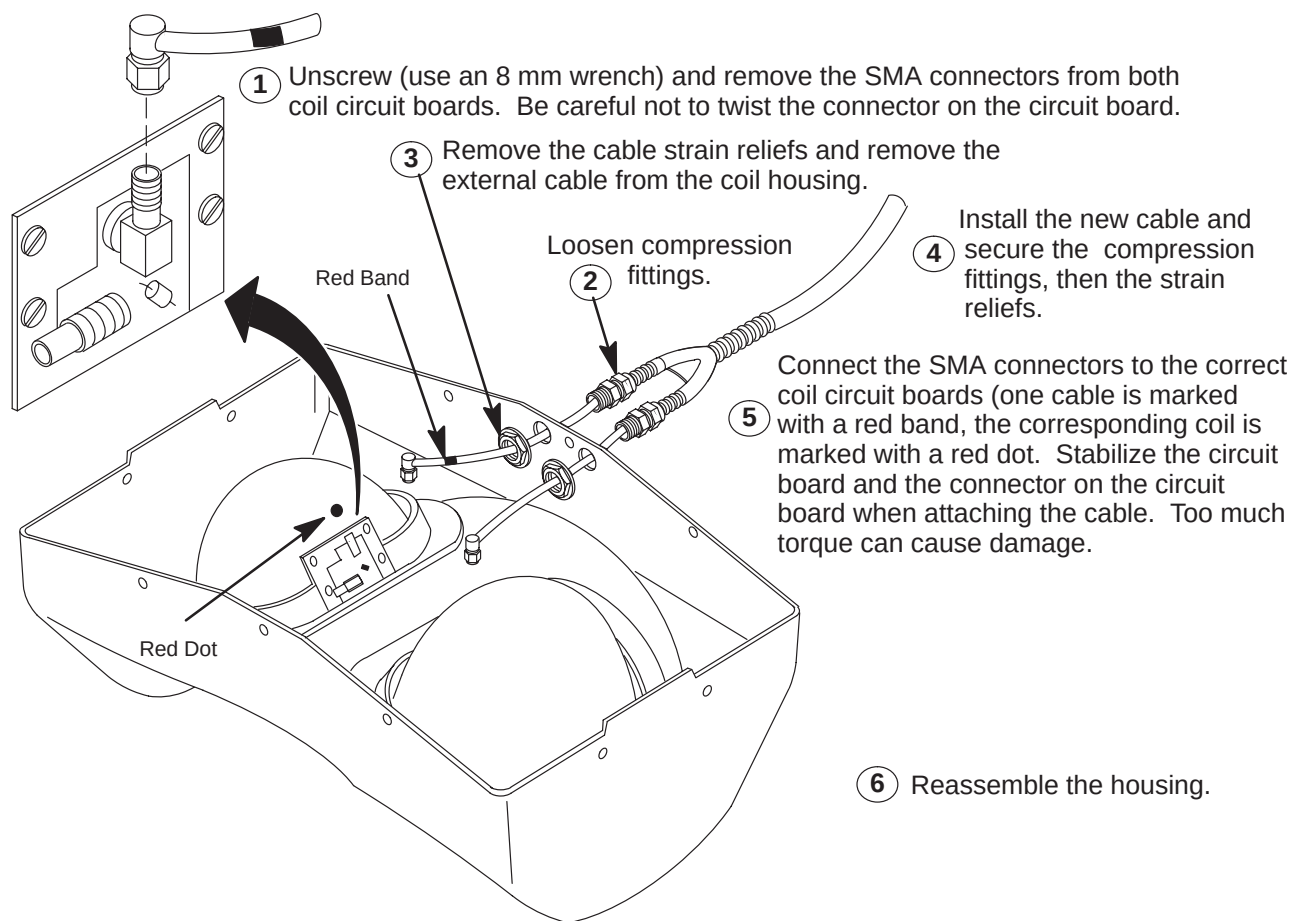
DIRECTION 46–305161

SECTION 4 – REPLACEMENT/MAINTENANCE

4-1 DISASSEMBLY/ASSEMBLY OF BREAST COIL



4-2 REPLACING THE EXTERNAL CABLE



4-3 REPLACING THE PROTECTION/DE-COUPLING PIN DIODES



Use a 25W solder iron and protect the circuit board from static electricity. Be careful not to damage the circuit board by overheating the solder.

Note

Each antenna loop element is tuned at the factory to provide the proper input impedance. Circuit boards are **not** field replaceable. The PIN diodes are the only field replaceable component on the circuit boards.

CB

L

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ACB

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PNW

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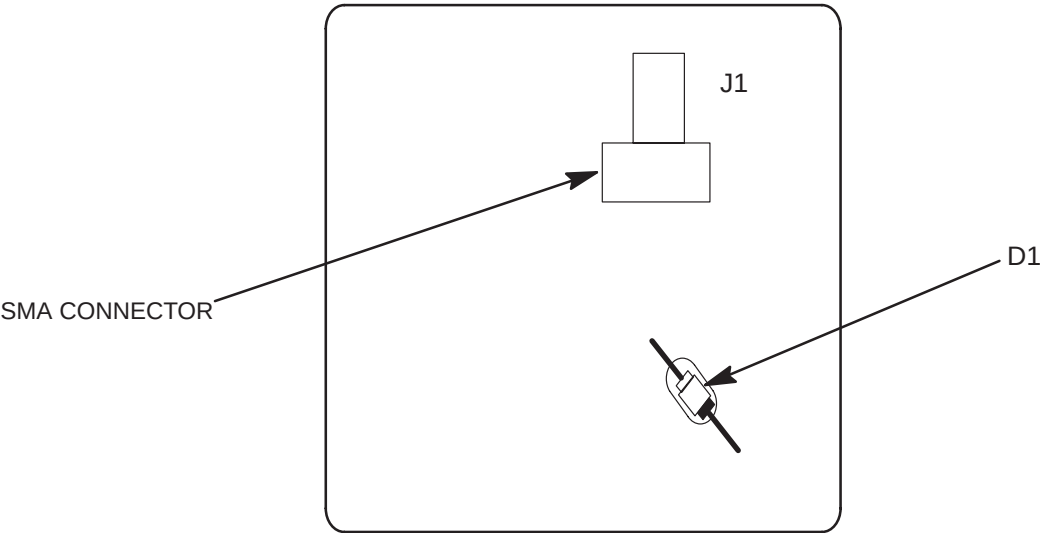
PU

BS

N

U

ACB



NOTE: NOT ALL COMPONENTS ARE SHOWN
COIL INPUT/DECOUPLING BOARD
ILLUSTRATION 4-1

4-4 CHECKING THE CABLES

Check the external cable for cracks or breaks once each week. Replace the external cable if any damage or wear is found. Refer to Section 4–2, REPLACING THE EXTERNAL CABLE, for external cable replacement procedure.

4-5 CLEANING THE BREAST COIL



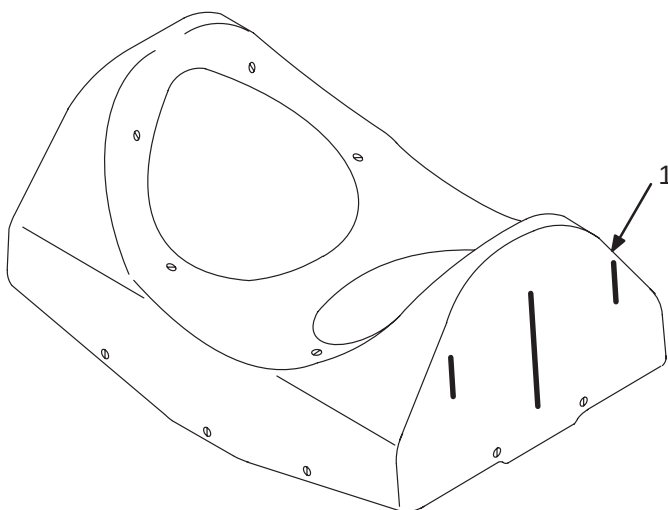
Avoid damaging sensitive electronic parts. Do not spray or pour dishwashing solution directly onto the Breast Coil, or external cable. Do not use alcohol to clean the Breast Coil. Never submerge the Breast Coil in any liquid.

Clean the Breast Coil and external cable with a mild dishwashing liquid and water solution. Wet a soft cloth with the solution and proceed to clean.

SECTION 5 – RENEWAL PARTS

BREAST COIL WITHOUT CABLE

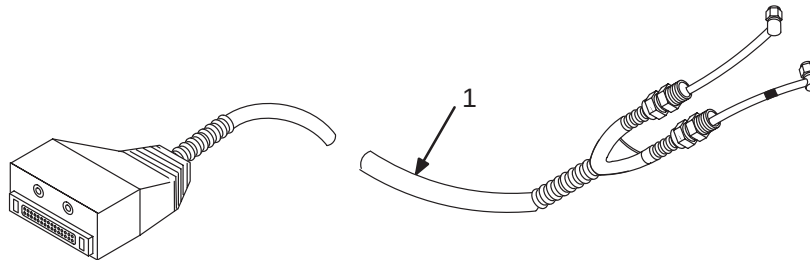
46-328288P1



Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	46-328288P1	1	BREAST COIL	1	BREAST COIL WITHOUT CABLE

PHASED ARRAY CABLE REPAIR KIT

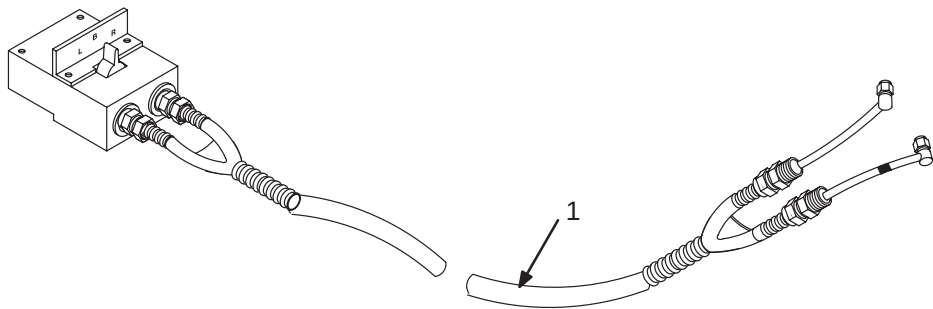
46-328289P1



Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	46-328289P1	1	CABLE KIT	1	PHASED ARRAY CABLE REPAIR KIT

SINGLE-RECEIVER CABLE REPAIR KIT

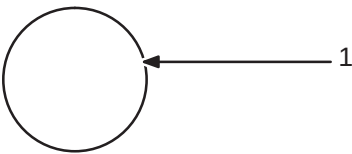
46-328290P1



Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	46-328290P1	1	CABLE KIT	1	SINGLE-RECEIVER CABLE REPAIR KIT

PHANTOM

46-317586G1



Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	46-317586G1	2	PHANTOM	1	100mm TLT PHANTOM

PIN DIODE

46-221735P1

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	46-221735P1	2	DIODE	1	UM9415 PIN DIODE

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DIRECTION 46-305161

DATA TABLE

Use the space provided below to record the calculated signal to noise ratio (SNR) data obtained from Section 3, FUNCTIONAL CHECKS. After recording the SNR data obtained during the initial coil installation, record subsequent SNR data in column 2 – 4 (for single-receiver coil) or column 5 (for phased array coil) below and the date they were obtained in column number 1 as a periodic QA check. If the ratio of any of the coils found is not greater than 85%, then there is a problem in the coil or the MR system.

Original SNR data obtained at initial coil installation:

Body Coil SNR Value: _____

Breast coil SNR Value: Unilateral Left _____ Both _____ Unilateral Right _____

OR Phased Array Coil _____

Date: _____

SNR DATA QA (QUALITY ASSURANCE) CHECK:

1	2	3	4	5	6	
Date:	Unilateral Left QA SNR Value:	Both QA SNR Value:	Unilateral Right QA SNR Value:	Phased Array Coil QA SNR Value:	Are new values divided by original values > 85%?	
_____	_____	_____	_____	_____	_____	Y/N
_____	_____	_____	_____	_____	_____	Y/N
_____	_____	_____	_____	_____	_____	Y/N
_____	_____	_____	_____	_____	_____	Y/N
_____	_____	_____	_____	_____	_____	Y/N
_____	_____	_____	_____	_____	_____	Y/N
_____	_____	_____	_____	_____	_____	Y/N
_____	_____	_____	_____	_____	_____	Y/N
_____	_____	_____	_____	_____	_____	Y/N
_____	_____	_____	_____	_____	_____	Y/N
_____	_____	_____	_____	_____	_____	Y/N
_____	_____	_____	_____	_____	_____	Y/N

Make additional copies of this document as is needed.

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DIRECTION 46-305161

DEFECTIVE COIL RETURN FORM

Note

To allow for proper assessment of defective returned coils, this form must be completely filled out and accompany all returned coils. Include films or prints of any image quality related complaints with a description of scan protocol used.

Date_____

Site Name_____

Site Address_____

Service Engineer_____

Coil Serial Number_____

Date Coil Installed_____

Description of Coil Problem_____

ELECTRICAL CHECKS

Vector Meter Checks – Coil Loaded With Human Subject In Free Space
Magnitude
Phase

Vector Meter Checks – Coil Unloaded In Free Space
Magnitude
Phase

Pin Diode Test
Diode Drop Forward Bias
Diode Drop Reverse Bias

Note

An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to "TLT PROCEDURE" calibration procedure located in Direction 15491, *Signa Advantage 1.5T & 0.5T System Troubleshooting Guide*. Use Table DR-1 and DR-2 to record the TLT results of the defective coil.

TABLE DR-1
TLT DATA FOR DEFECTIVE PHASED ARRAY COIL

SITE:		NAME:		DATE:	
TLT FILE NUMBER:					
SNR:	NOISE (0)				
	NOISE (1)				
	NOISE (2)				
	NOISE (3)				
	NOISE (C)				
	SNR MEAN (0)				
	SNR MEAN (1)				
	SNR MEAN (2)				
	SNR MEAN (3)				
	SNR MEAN (C)				
TR MAP: (Combined Results)	89-91 (C)		%		%
	85-95 (C)		%		%
	65-115 (C)		%		%
	FLIP MEAN (C)				
	FLIP SDV (C)				
	RCV MEAN (C)				
	RCV SDV (C)				
	DIFF MEAN (C)				
	DIFF SDV (C)				

TABLE DR-2
TLT DATA FOR DEFECTIVE SURFACE COIL

SITE:		NAME:		DATE:	
TLT FILE NUMBER:					
SNR:	NOISE				
	SNR MEAN				
	SNR SIGNAL				
	SNR AREA				
TR MAP:	89-91		%		%
	85-95		%		%
	65-115		%		%
	FLIP MEAN				
	FLIP SDV				
	RCV MEAN				
	RCV SDV				
	DIFF MEAN				
	DIFF SDV				